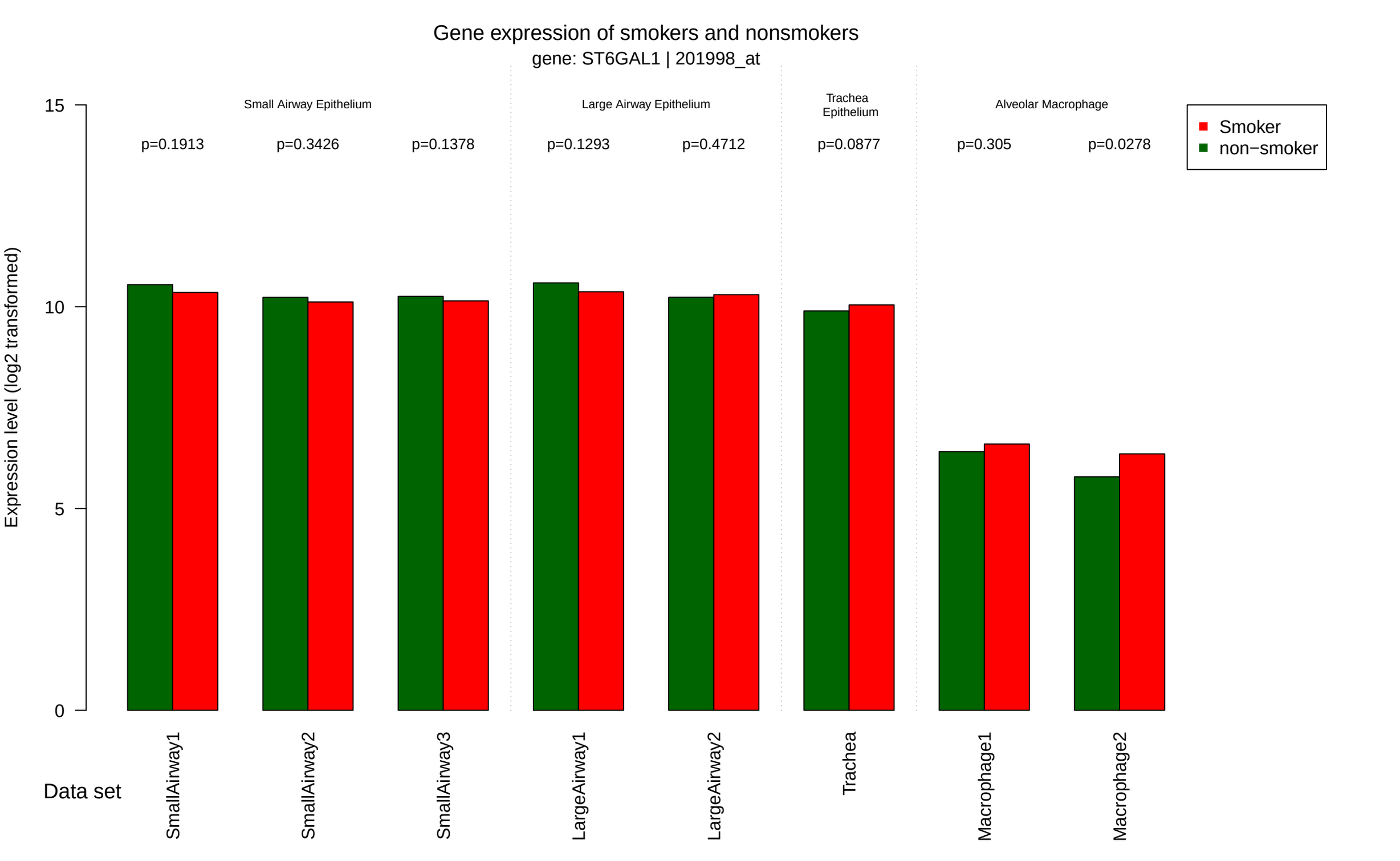
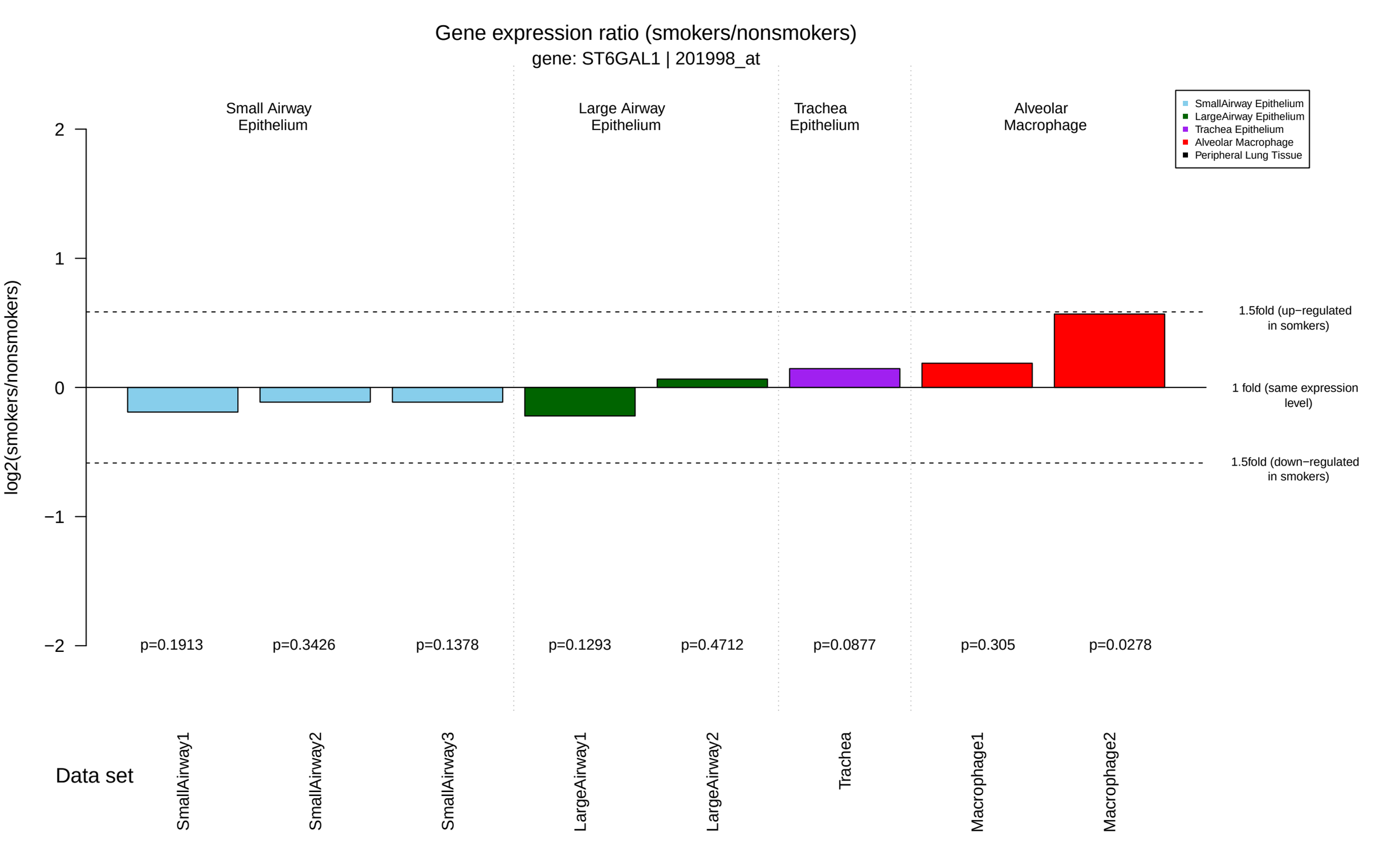
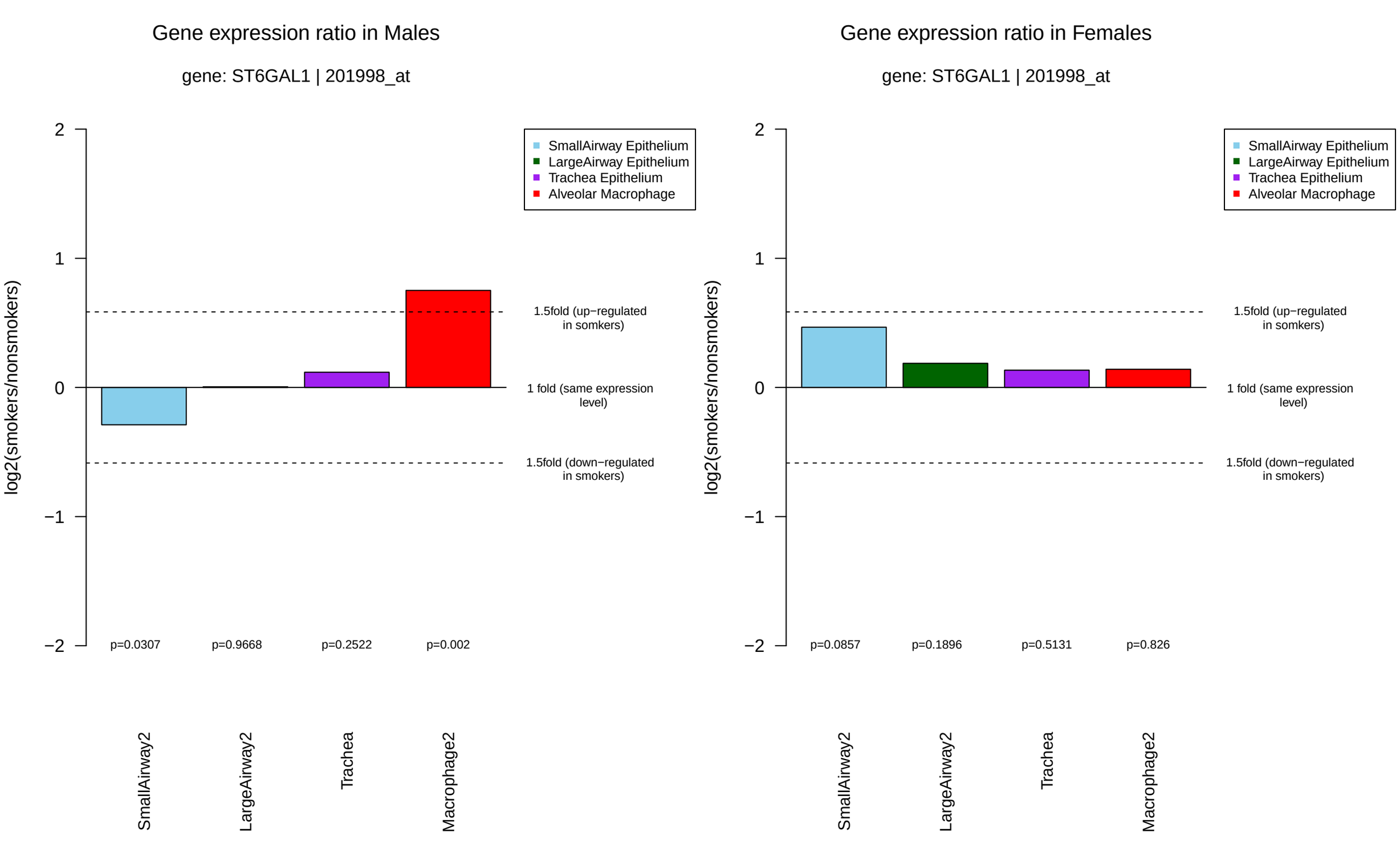
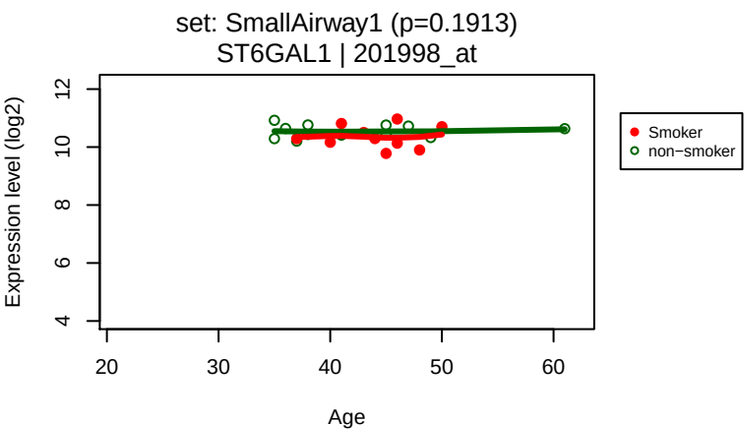
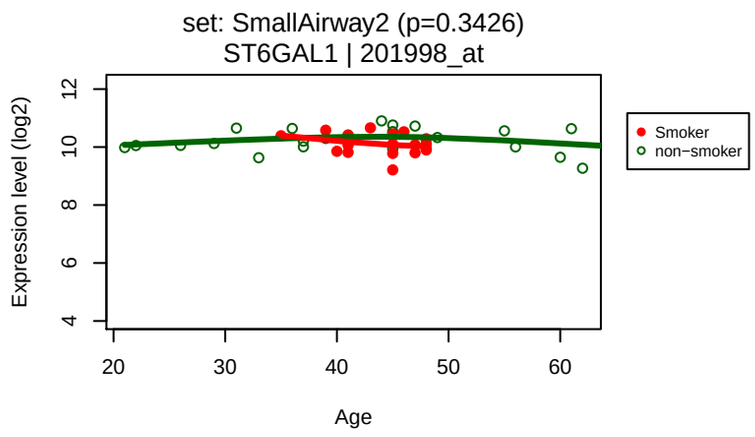
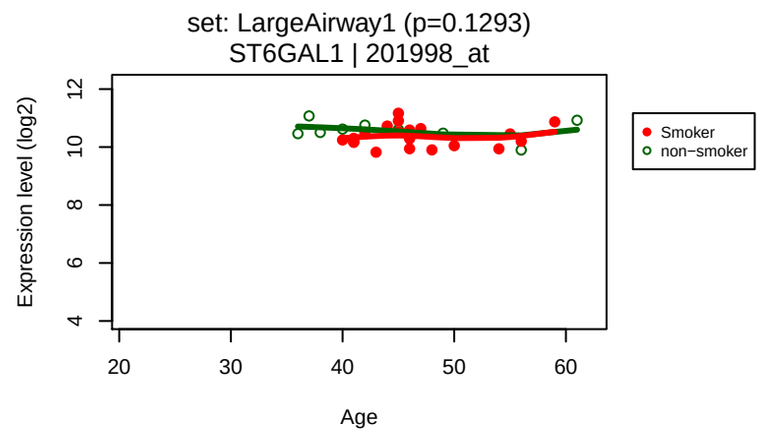

Gene expression of smokers and nonsmokers
gene: ST6GAL1 | 201998_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1913
p=0.3426
p=0.1378
p=0.1293
p=0.4712
p=0.0877
p=0.305
p=0.0278
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

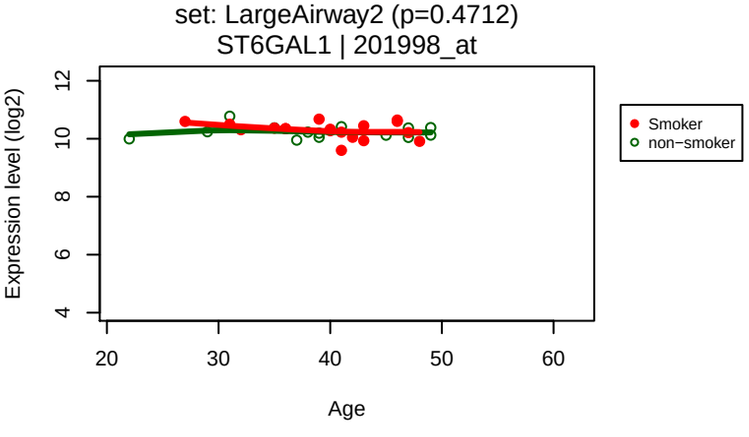

Gene expression ratio (smokers/nonsmokers)
gene: ST6GAL1 | 201998_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1913
p=0.3426
p=0.1378
p=0.1293
p=0.4712
p=0.0877
p=0.305
p=0.0278
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

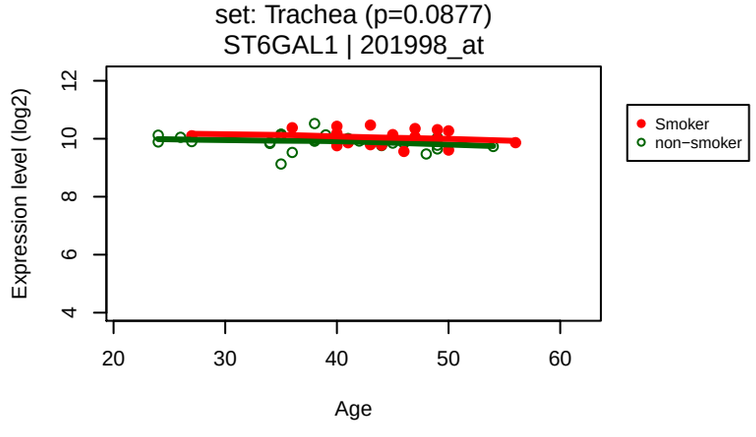

Gene expression ratio in Males
gene: ST6GAL1 | 201998_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0307
p=0.9668
p=0.2522
p=0.002
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GAL1 | 201998_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0857
p=0.1896
p=0.5131
p=0.826
SmallAirway2
LargeAirway2
Trachea
Macrophage2

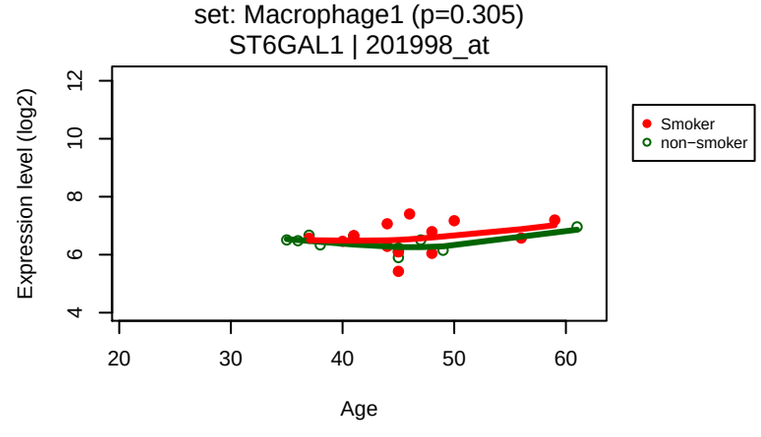

set: SmallAirway1 (p=0.1913)
ST6GAL1 | 201998_at
Expression level (log2)
Age
Smoker
non−smoker

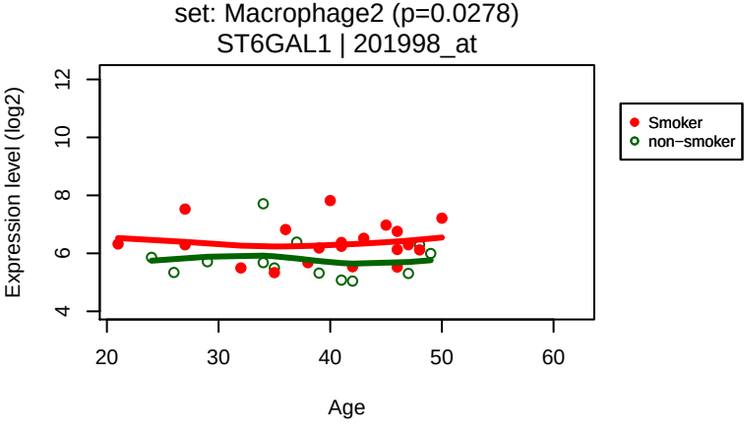

set: SmallAirway2 (p=0.3426)
ST6GAL1 | 201998_at
Expression level (log2)
Age
Smoker
non−smoker

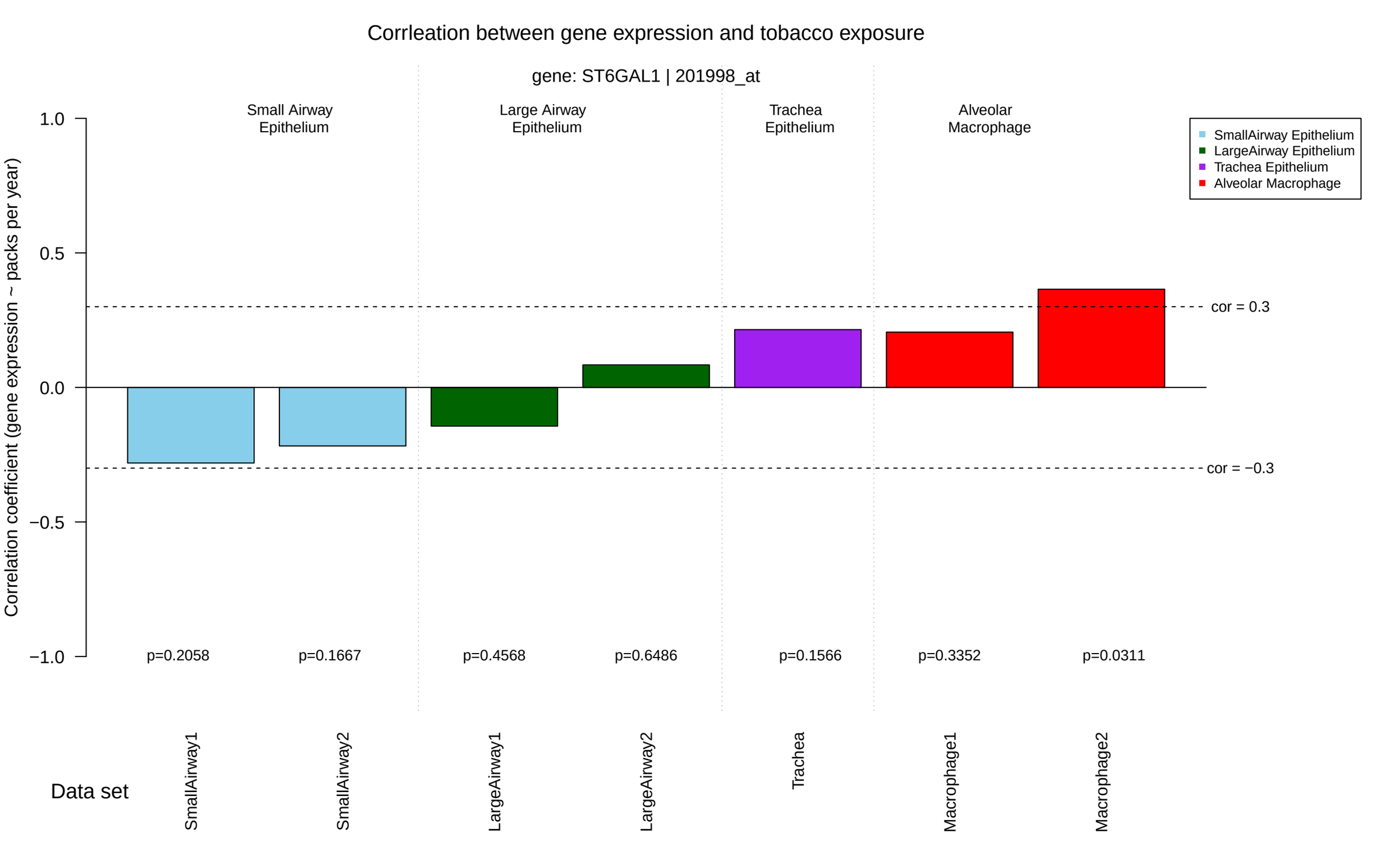

set: LargeAirway1 (p=0.1293)
ST6GAL1 | 201998_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4712)
ST6GAL1 | 201998_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0877)
ST6GAL1 | 201998_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.305)
ST6GAL1 | 201998_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0278)
ST6GAL1 | 201998_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GAL1 | 201998_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2058
p=0.1667
p=0.4568
p=0.6486
p=0.1566
p=0.3352
p=0.0311
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2